



Alberta Public Health Disease Management Guidelines

Paratyphoid Fever



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Health and Wellness Promotion Branch

Public Health and Compliance Branch

Alberta Health

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Table of Contents

Case Definition	4
Confirmed Case	4
Probable Case	4
Carrier	4
Reporting Requirements	5
Physicians, Health Practitioners and Others	5
Laboratories	5
Alberta Health Services and First Nations and Inuit Health Branch	5
Epidemiology.....	6
Etiology	6
Clinical Presentation.....	6
Diagnosis	6
Treatment.....	6
Transmission.....	6
Incubation Period	6
Period of Communicability.....	6
Host Susceptibility	7
Incidence.....	7
Public Health Management.....	8
Key Investigation.....	8
Management of a Case.....	9
Management of Contacts	10
Management of Chronic Carriers and Their Contacts	11
Preventive Measures.....	11
Appendix 1: Cases Involved in Sensitive Situations or Occupations	12
Appendix 2: Contacts Involved in Sensitive Situations or Occupations.....	13
Appendix 3: Revision History	14
References.....	15

Case Definition

Confirmed Case

Laboratory confirmation of infection with or without clinical illness^(A):

- Isolation of *Salmonella paratyphi* A, B, or C from an appropriate clinical specimen (e.g., sterile site, deep tissue wounds, stool, vomit or urine)^(B).

Probable Case

Clinical illness^(A) in a person who is epidemiologically linked to a confirmed case.

Carrier

Individuals who continue to shed *Salmonella paratyphi* for one year or greater are considered to be carriers^(C).

NOTE: *Salmonella paratyphi B var java* is considered a case of Salmonella (i.e., non-typhoidal) and should not be reported as Paratyphoid Fever.

^(A) Clinical illness is characterized by headache, diarrhea, abdominal pain, nausea, fever and sometimes vomiting. Asymptomatic infections may occur, and the organism may cause extra-intestinal infections.

^(B) Refer to the current [Public Health Laboratories \(ProvLab\) Guide to Services](#) for specimen collection and submission information.

^(C) Alberta Health maintains a Typhoid/Paratyphoid Registry for purposes of monitoring carriers as they potentially pose a long term health risk for transmission of disease.

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed and probable cases in the prescribed form by the Fastest Means Possible (FMP).

Laboratories

All laboratories shall report all positive laboratory results:

- by FMP to the MOH (or designate) of the zone, and
- by mail, fax or electronic transfer within 48 hours (two business days) to the Chief Medical Officer of Health (CMOH) (or designate).

Alberta Health Services and First Nations and Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall notify the CMOH (or designate) by FMP of all confirmed and probable cases.
- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and probable cases to the CMOH (or designate) within one week of notification and the final NDR (amendment) within two weeks of notification.
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by FMP:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Epidemiology

Etiology

Paratyphoid fever is caused by *Salmonella enterica* subsp. *enterica* serotype *S. paratyphi* A, *S. paratyphi* B (*S. schottmuelleri*) and *S. paratyphi* C (*S. hirschfeldii*).⁽¹⁾

Clinical Presentation

Paratyphoid is a systemic bacterial disease that presents with clinical symptoms similar to typhoid but generally less severe and shorter in duration. Symptoms include an insidious onset of fever, headache, malaise, anorexia, a non-productive cough in the early stages of the illness, a relative bradycardia and splenomegaly.⁽¹⁾ A transient, macular rash of rose-colored spots can occasionally be seen on the trunk.⁽²⁾ In adults, constipation is more often seen than diarrhea. Relapses can occur in approximately 3–4% of cases.

A carrier state may follow acute or mild illness and even subclinical infections. Persons with *S. Paratyphi* infections tend to become carriers less frequently than persons infected with *S. typhi*.⁽¹⁾

Diagnosis

The causative organisms can be isolated from the blood early in the disease, and from urine and feces after the first week. Bone marrow culture provides a greater sensitivity for bacteriological confirmation even in persons who have already received antibiotics.⁽¹⁾

Treatment

Generally fluoroquinolones are effective for treatment in adults. However, antibiotic resistance has become a public health problem in many areas (especially Asia) and therefore, clinicians should consider local resistance patterns when choosing appropriate antimicrobials for treatment.^(1,7,8) Consulting with an infectious diseases specialist may be appropriate.

Transmission

Paratyphoid is transmitted through ingestion of food and/or water contaminated by urine or feces from infected cases or carriers. The infection is rarely spread by casual contact. Shellfish (particularly oysters) taken from sewage-contaminated beds; fruits and/or vegetables fertilized by night soil (human excrement) eaten raw; and contaminated milk/milk products (usually contaminated by hands of carriers) are important sources of infection to consider. Flies may also infect foods, whereby the organism can multiply to achieve an infective dose.⁽¹⁾

Incubation Period

The incubation period ranges from 1–10 days depending on the size of infecting dose and host factors.⁽¹⁾

Period of Communicability

The period of communicability lasts as long as the bacilli are present in the excreta. This usually begins from the first week after onset of illness and continues through convalescence and for a variable period thereafter (commonly one to two weeks for paratyphoid). Fewer people infected with paratyphoid fever become permanent carriers compared to those with typhoid fever.⁽¹⁾

Host Susceptibility

Susceptibility is general and is increased in persons with achlorhydria (a condition in which production of gastric acid in the stomach is absent or low, or more commonly treatment with acid suppression agents) (S. Houston, 2013, Personal Communication, January 28). Relative specific immunity follows recovery from clinical illness or inapparent infection.⁽¹⁾

Incidence

General

Worldwide occurrence. An estimated six million cases of paratyphoid fever occur worldwide annually and approximately 150 cases of paratyphoid fever are reported each year in the United States, most of which are in recent travellers. The risk of paratyphoid fever is increasing among travellers to southern and Southeast Asia.⁽²⁾ Paratyphoid fever occurs sporadically or in limited outbreaks but is most likely under reported.⁽¹⁾ The ratio of disease caused by *S. typhi* compared to that of *S. paratyphi* A and B is estimated to be about 4:1.⁽¹⁾ Parts of China and Pakistan are reporting more cases caused by *S. paratyphi* than *S. typhi*.⁽¹⁾

Canada

The incidence of paratyphoid remains very low in Canada. In 1999, *S. paratyphi* infection was removed from the nationally notifiable disease list as a separate entity and was included under Salmonella infections. Laboratory confirmed isolations of enteric pathogens in Canada, including bacterial, viral and parasitic pathogens are reported on a weekly basis through the National Enteric Surveillance Program (NESP). Between 2006 and 2011 there was an average of 102 *S. paratyphi* isolates reported to the NESP per year. The majority of isolates were identified as *S. paratyphi* A.⁽³⁾

Alberta

The greatest risk of paratyphoid infection for Albertans appears to be travel to countries where sanitation is likely to be poor or sewage systems inadequate. Between 2000 and 2012, the number of cases of paratyphoid reported annually varied from two (2003) to 13 (2011) representing rates of 0.06 cases per 100,000 to 0.34 cases per 100,000 respectively. During this same period of time, 93/104 cases reported travel outside Canada (**e.g.**, travelled to South Asia), an additional 6/104 cases reported recent immigration and for 5/104 cases there was no information available.

Public Health Management

Key Investigation

- Confirm the diagnosis.
- Obtain a history of illness including the date of onset, signs and symptoms.
- Identify any underlying medical conditions (e.g., decreased gastric acidity).
- Determine the occupation of the case (e.g., food handler, childcare facility worker, health care worker) and identify specific duties at work.
- If the case is a child, determine attendance at a childcare facility (e.g., daycare, dayhome) or other childcare site.
- Determine whether the case had any other type of institutional contact (e.g., long term care).
- Determine the possible source of infection taking into account the incubation period, reservoir, and mode of transmission. Assessment should include the following.
 - Determine history of travel. **NOTE:** Additional specimens (e.g., urine for *S. paratyphi* +/- serology for schistosomiasis) may be indicated in special circumstances. Refer to [Management of a Case](#).
 - Determine history of residing in areas with poor sanitation, including improper water treatment and sewage disposal (including recent immigration).
 - Obtain a detailed food history (including consumption of shellfish).
 - Identify any risk behaviours including lifestyle risks for infection (e.g., high risk sexual practices, especially contact with feces).
 - Determine any exposure (either household or non-household) to a confirmed case of paratyphoid.
 - Identify symptomatic household or other close contacts (e.g., co-travellers or others) who have recently travelled to a developing country.
- Identify contacts that may have had exposure during the period that the case was infectious, especially those that are in sensitive situations or occupations (see Table 1). Consider the following individuals when identifying contacts:
 - household contacts,
 - recent co-travellers (includes anyone who travelled with the case and is likely to have been exposed to the same source of infection as the case rather than someone who only travelled on the same plane/bus as the case), or
 - other close, non-household contacts (e.g., sexual contacts, childcare site contacts).

Table 1: Sensitive Situations or Occupations (SSO)

SSO	Definition
Food handler	• Touches unwrapped food to be consumed, <u>and/or</u> • Handles equipment or utensils that touch unwrapped food to be consumed.*
Healthcare, child care or other staff	• Has contact through serving food to highly susceptible persons. • Provides direct patient care and is involved in the care of young children, elderly or dependent persons.
Child attending a childcare facility or similar facilities	• Is diapered or unable to implement good standards of personal hygiene.
Any individual (older child or adult)	• Is unable to implement good standards of personal hygiene (e.g., with disabilities/challenges that may impact ability to perform good hand hygiene) and is involved in an activity that may promote disease transmission.

* NOTE: Generally, food handlers who do not touch food, equipment or utensils in this way are not considered to pose a transmission risk; however, circumstances for each case should be assessed on an individual basis.

Management of a Case

- Provide information about disease transmission and the appropriate infection prevention and control measures to be implemented to minimize the possibility of transmission including strict hand hygiene especially:
 - after using the washroom,
 - after changing diapers, and
 - before preparing/handling and serving food.
- Advise the case to refrain from preparing food for others for the duration of the period of communicability.
- Advise the case to remain off work until they are free from diarrhea and other gastrointestinal symptoms and for 48 hours following resumption of normal stool.
 - Continued public health surveillance is not required for these cases, however, follow-up with the personal physician for clearance of the organism is recommended. Persons not involved in SSO present a minimal risk of spreading gastrointestinal pathogens if they are healthy and have normal, well formed stools.

Hospitalized Patients

- Routine practices should be followed.
- For hospitalized children or adults with poor hygiene habits or who have incontinence that cannot be contained, additional precautions (i.e., contact precautions) should be implemented.⁽⁵⁾

Exclusion – SSO

- Exclude symptomatic and asymptomatic cases who are involved in SSO (see [Table 1](#)).
- Exclusion applies until two consecutive stool specimens collected from a confirmed case are reported as negative **AND** one urine culture for *S. paratyphi* is reported negative if case has EVER travelled to or lived in a schistosomiasis endemic country and may have been exposed to schistosomiasis.
 - Collect stool specimens after stools have returned to normal and at least 21 days following completion of antibiotic and not less than 24 hours apart.
 - If one or both stool samples are positive for *S. paratyphi*, continue submitting samples for testing at the specified intervals as outlined in [Appendix 1](#).
 - If urine sample is positive for *S. paratyphi*, collect serology for schistosomiasis:
 - eight weeks post exposure in areas where risk exists for infection with *Schistosoma japonicum*, *S. mansoni*, *S. mekongi* and *S. intercalatum*, or
 - twelve weeks post exposure in areas where risk exists for infection with *S. haematobium*.

Specimens collected before this time, may yield false negative results (Kowaleska-Grochowska, K., 2012, Personal Communication, Nov 19). See [Appendix 1](#).

 - If urine sample is positive for *S. paratyphi* and serology is positive for schistosomiasis, refer the case back to physician and advise physician to treat case concurrently for both infections, even if this means repeating antibiotic treatment. Refer to [Appendix 1](#).
 - Persons co-infected with schistosomiasis should be treated with paraziquantel to eliminate possible carriage of salmonella bacteria (including *S. typhi* and *S. paratyphi*) by the schistosomes.⁽¹⁾
- If possible, consideration should be given to temporary redeployment away from activities that involve increased risk of transmission.

Management of Contacts

- “Warn and inform” contacts by providing information about disease transmission and appropriate infection prevention and control measures. Stress the measures that need to be taken to minimize possible fecal-oral transmission including strict hand hygiene, especially after using the washroom, changing diapers, and before eating and preparing/handling foods.
- Refer all symptomatic contacts to their physician for assessment.
- Refer asymptomatic co-travellers who are not involved in SSO to their own health care provider for screening and follow-up.
- Public health follow-up is generally not required for contacts of a case who are not involved in SSO; however, the circumstances should be considered individually. Follow-up with their health care provider is appropriate.
- Additional contact follow-up and assessment of a source of infection may be necessary if cases are identified with no recent travel history and/or for whom travel is an unlikely source of infection. Stool samples may be requested on identified contacts to determine the source of the infection in the case.

Exclusions – SSO

- See [Appendix 2](#) for information on stool collection for contacts involved in SSO.
- Exclude **symptomatic contacts** involved in SSO until two consecutive stool specimens are taken not less than 24 hours apart and are reported as negative prior to returning to work or back to their childcare facility.
 - If any stool sample is positive, manage as a confirmed case.
 - If stool sample is negative, advise contact to:
 - remain home while symptomatic and until 48 hours after last symptoms before returning to work or childcare facility,
 - continue to self-monitor for symptoms within 20 days after initial negative stools, and
 - seek prompt medical attention and inform public health if symptoms develop.
- Screen **asymptomatic co-travellers** who are involved in SSO:
 - Collect one stool specimen and question about symptoms both while away travelling and since returning.
 - No exclusion is necessary for asymptomatic co-travellers while waiting for results of screening stool specimen. If stool result is positive, they would then be treated and managed as a confirmed case.
- Collect ONE screening stool for **asymptomatic household or other close, non-household contacts (e.g., sexual contacts)** involved in SSO and **asymptomatic childcare site contacts (e.g., staff and children in the same room)**.
 - Normally exclusion is not necessary while waiting for results of screening.
 - If the individual is unable/unwilling to comply with good hand hygiene OR unwilling to provide a screening stool sample, exclusion is **necessary** prior to returning to work or back to their childcare facility until one stool specimen is submitted and reported as negative.
 - Consultation with the MOH is appropriate.
 - Contacts should be advised to self-monitor and seek prompt medical attention AND contact public health if they develop symptoms within 20 days of the initial negative stool result.

Management of Chronic Carriers and Their Contacts

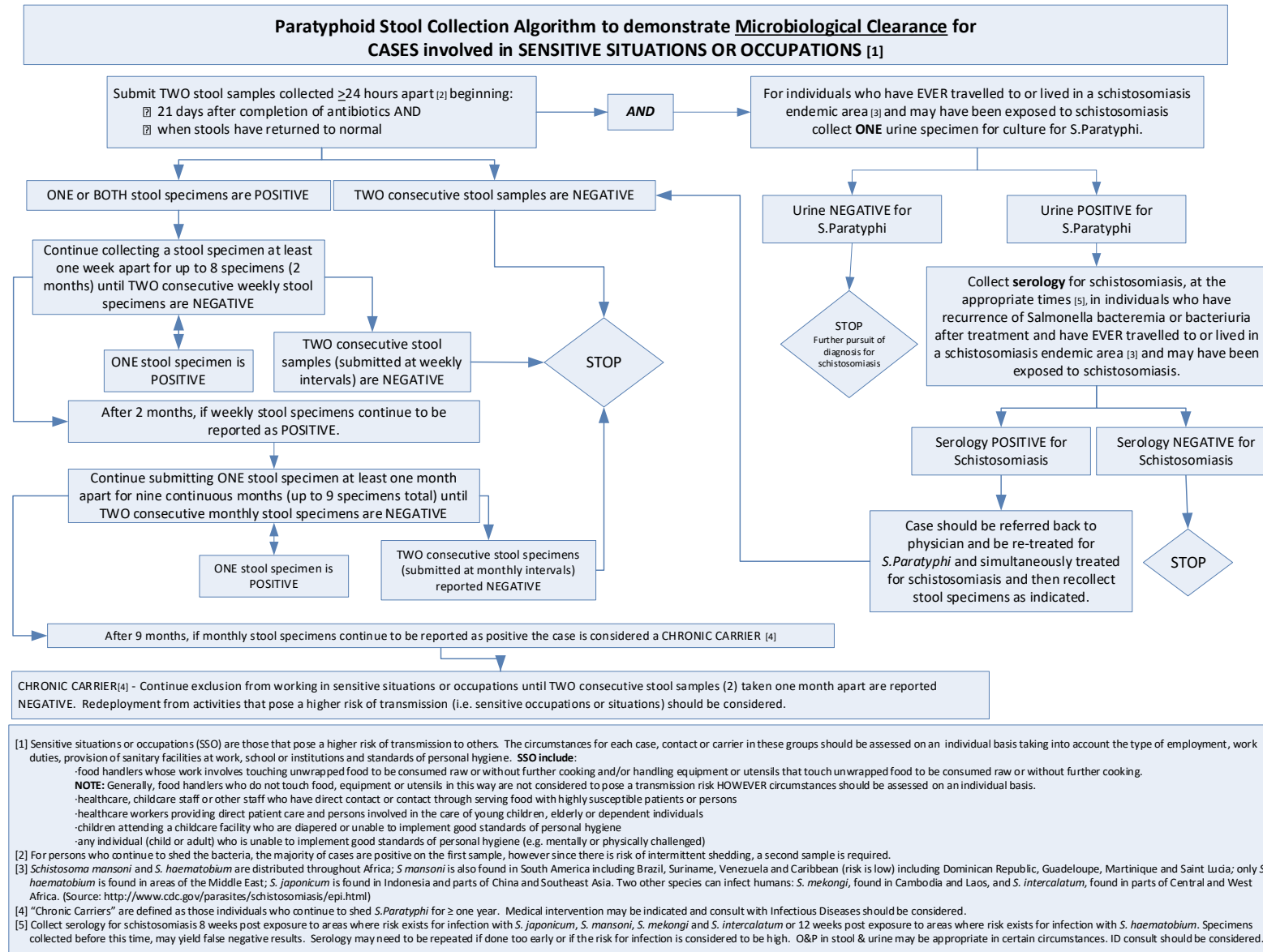
Individuals who continue to shed *S. paratyphi* for one year or greater are considered carriers.

- Refer to [Appendix 1](#) for details on continued stool testing for carriers.
- Medical intervention may be indicated for individuals who become carriers and consultation with an infectious disease specialist should be considered.
- Exclude chronic carriers from activities that are involved in SSO and continue submitting stool specimens as outlined in [Appendix 1](#).
 - If possible, consideration may be given to temporary redeployment of the carrier away from activities that involve increased risk of transmission.
- **Contacts of carriers** should be advised to seek prompt medical assessment and screening and notify public health if they become symptomatic.
 - Exclude symptomatic contacts of carriers who are involved in SSO until two consecutive stool specimens taken not less than 24 hours apart are reported as negative prior to returning to work or back to their childcare facility.
 - Asymptomatic contacts of carriers are not excluded and no stool specimens are required.
 - Public health follow-up is generally not required for contacts of a carrier who are not involved in SSO; however, stool samples may be requested to determine the source of the infection in the case.

Preventive Measures

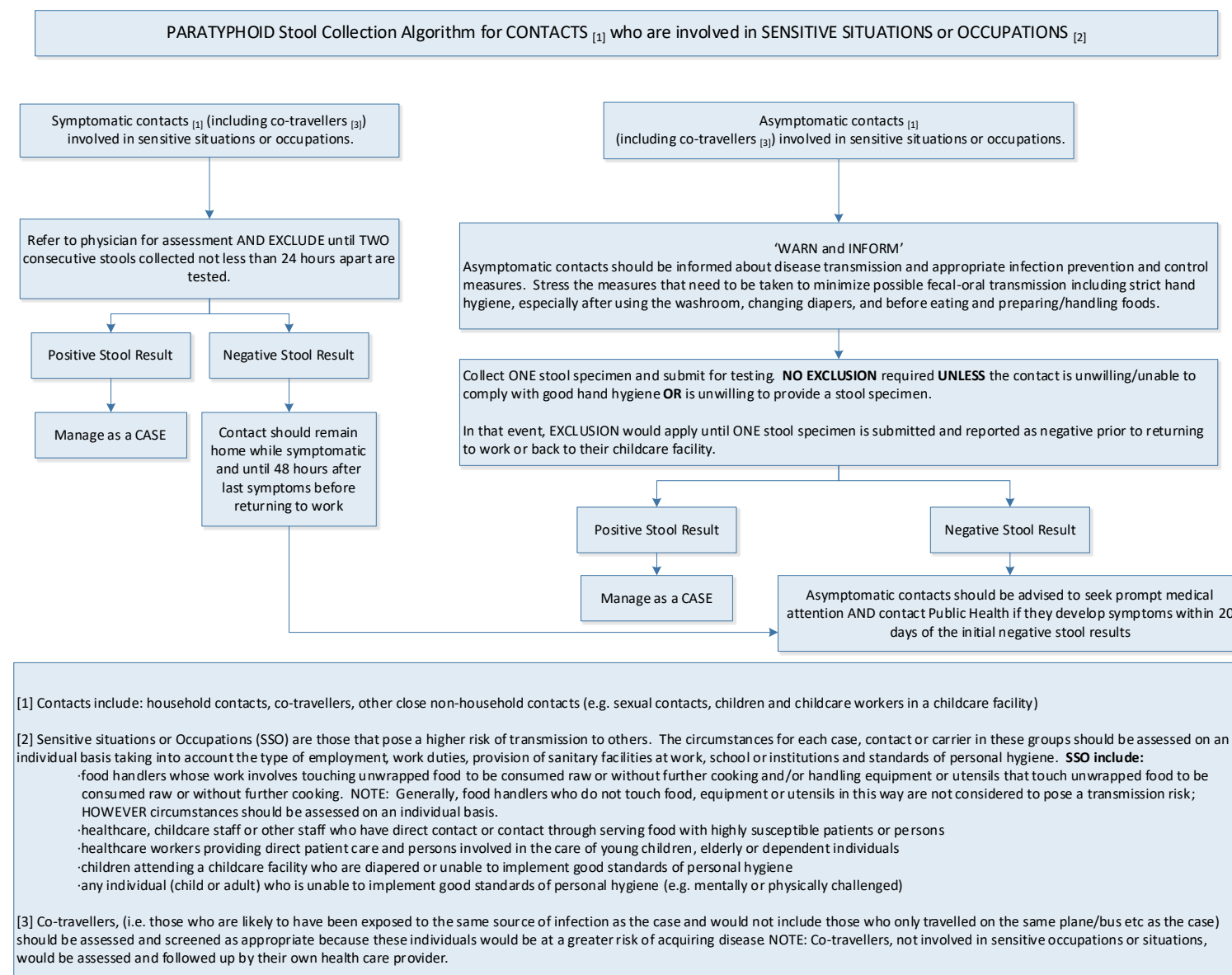
- There is currently no vaccine to protect against paratyphoid fever.
- Educate the public regarding the need for:
 - thorough hand washing especially after using the washroom, changing diapers, and before eating and preparing/handling foods,
 - sanitary disposal of feces, and
 - hygienic food preparation and food and equipment handling practices.
- Reinforce information about boiling or steaming shellfish for at least 10 minutes before serving.
- Encourage all travellers to seek travel advice before visiting areas of high prevalence of disease. Emphasize the importance of good sanitation, proper arrangements for safe water supplies and scrupulous personal hygiene while travelling.

Appendix 1: Cases Involved in Sensitive Situations or Occupations



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Appendix 2: Contacts Involved in Sensitive Situations or Occupations



Appendix 3: Revision History

Revision Date	Document Section	Description of Revision
October 2021	General	• Updated Template • Treatment section moved to Epidemiology • Updated web links
	Public Health Management	• Terminology changed from Sensitive Occupations or Situations to Sensitive Situations or Occupations to align with other enteric diseases • Layout and minor wording changes to improve clarity

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